

1. Conceptual Data Model

- Entities:
- Customer
- Order
- OrderSummary
- CustomerAggregateSpend
- Relationships:
- Customer places Order (1-to-many: Customer → Order)
- OrderSummary is derived by joining Customer and Order, tracking SCD Type 2 history
- CustomerAggregateSpend aggregates OrderSummary by customer and date
- Entity Definitions:
- Customer: Contains customer master data.
- Order: Represents purchases made by customers.
- OrderSummary: Combines customer and order data, tracks historical changes.
- CustomerAggregateSpend: Aggregates each customer's total daily spending.

2. Logical Data Model

- Customer
- CustId (Primary Key)
- Name
- EmailId
- Region
- Order
- OrderId (Primary Key)
- ItemName
- PricePerUnit
- Qty
- Date
- CustId (Foreign Key → Customer.CustId)
- TotalAmount (Computed: PricePerUnit × Qty)
- OrderSummary
- CustId
- Name

- EmailId
- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date
- StartDate (SCD2 Field,)
- EndDate (SCD2 Field,)
- Active (SCD2 Field,)
- Composite Key: CustId, OrderId, StartDate
- FKs: CustId (to Customer), OrderId (to Order)
- CustomerAggregateSpend
- Name
- TotalAmount (SUM aggregated per name/date)
- Date
- Composite Key: Name, Date
- Normalization:
- All tables are in 3NF.
- Each non-key attribute depends on the full primary key.
- Business Rules by Entity:
- Remove "Null"/Null literals and duplicates (Customer, Order)
- Compute TotalAmount in Order.
- Use SCD Type 2 tracking in OrderSummary.
- Aggregate spend by Name and Date in CustomerAggregateSpend.

3. Physical Data Model

- Table: customer
- Columns: CustId, Name, EmailId, Region
- Datatypes:
- PK: CustId
- Table: order
- Columns: OrderId, ItemName, PricePerUnit, Qty, Date, CustId, TotalAmount
- Datatypes:

- PK: OrderId
- FK: CustId (to customer)
- Table: ordersummary
- Columns: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date, StartDate, EndDate, Active
- Datatypes:
- Composite PK: CustId, OrderId, StartDate
- FKs: CustId (to customer), OrderId (to order)
- Table: customeraggregatespend
- Columns: Name, TotalAmount, Date
- Datatypes:
- Composite PK: Name, Date
- Indexing & Partitioning:
- Indexes on keys.
- Partition by Date where relevant.
- Storage: Distributed, enterprise data lake/lakehouse.

4. Naming Conventions

- Table names: Lowercase, underscores (e.g., customeraggregatespend)
- Column names: As in requirements, preserve casing.
- PK/FK conventions: pk_tablename_col, fk_tablename_refcol

5. Data Dictionary

Entity	Field	Definition	Notes
Customer	CustId	Unique Customer Identifier	Primary Key
Customer	Name	Customer Name	
Customer	EmailId	Email Address	
Customer	Region	Customer Region	
Order	OrderId	Order Identifier	Primary Key
Order	ItemName	Item Purchased	
Order	PricePerUnit	Price per unit	
Order	Qty	Quantity	
Order	Date	Order Date	
Order	CustId	Customer Identifier (FK)	FK: Customer
Order	TotalAmount	Computed Order Total	PricePerUnit × Qty

OrderSummary	CustId	Customer Identifier	PK/FK
OrderSummary	Name	Customer Name	
OrderSummary	EmailId	Email Address	
OrderSummary	Region	Customer Region	
OrderSummary	OrderId	Order Identifier	PK/FK
OrderSummary	ItemName	Item Purchased	
OrderSummary	PricePerUnit	Price per unit	
OrderSummary	Qty	Quantity	
OrderSummary	Date	Transaction Date	
OrderSummary	StartDate	SCD2 Start Date	<TBD>
OrderSummary	EndDate	SCD2 End Date	<TBD>
OrderSummary	Active	SCD2 Active Flag	<TBD>
CustomerAggregateSpend	Name	Customer Name	Aggregation key
CustomerAggregateSpend	TotalAmount	Aggregated daily spend	SUM
CustomerAggregateSpend	Date	Spend Date	Aggregation key

6. ERD Description

[Customer]

*CustId

Name

EmailId

Region

[Order]

*OrderId

ItemName

PricePerUnit

Qty

Date

+CustId

TotalAmount

[OrderSummary]

*CustId

*OrderId

Name

EmailId

Region

ItemName

PricePerUnit

Qty

Date

StartDate

EndDate

Active

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7. Open Questions

- Data types and nullability for all columns ()
- Explicit SCD2 tracking field specs ()
- Partitioning, clustering, storage optimization ()
- Operational and performance parameters ()